Landslide Early Warning System (Kathmandu- Naranghath Corridor)

Introduction

The steep, monsoon prone highway between Naranghath and Kathmandu is a critical transport link i.e this is the only highway that connects western Nepal and Terai to Kathmandu, but this highway suffers frequent landslide disruption [1][2]. For example, heavy rains in summer 2023 forced nine closures of this highway [1]. In collaboration with the University of the Sunshine Coast (Australia), Nepal's Disaster Risk Reduction Authority (NDRRMA), and a Nepalese geospatial partner, we propose an automated system that fuses Sentinel-1 InSAR displacement with DHM rainfall forecasts to trigger timely landslide alerts. NDRRMA – Nepal's designated DRR authority – serves as secretariat to the national DRR council and will coordinate this early warning system EWS effort [3][4].

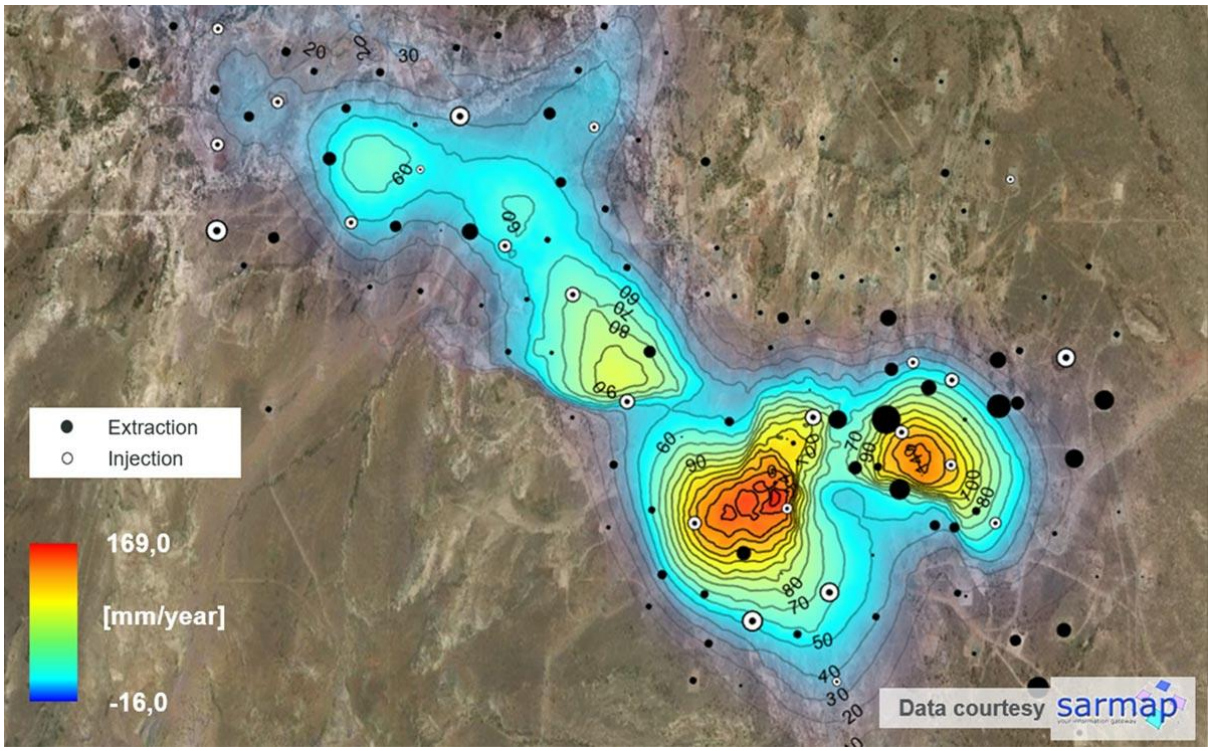
Fig1: Landslide in Naranghath Kathmandu route

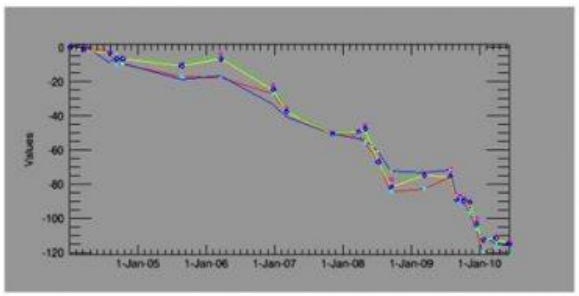
Stakeholders and Goals:

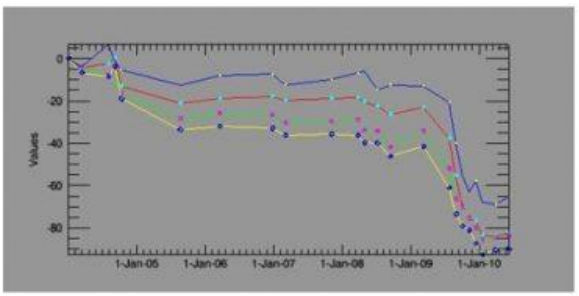
APAC geospatial Nepal team is responsible for InSAR research and system architecture design and development, but the University expertise will lead this project. NDRRMA (with District DRR committees) defines operational requirements and ensures warnings align with national DRR plans[3][4]. All parties aim to enhance Nepal's multi-hazard EWS (per Sendai/UN initiatives) by adding a new, data-driven landslide component[3][5]. Specifically, we align with NDRRMA's mandate to integrate landslide forecasts into existing disaster management policies[4][3].

Datasets:

We will use free, all-weather Sentinel-1 SAR data (C-band) which deliver global coverage every 6 days[6] (24-hour data latency) and are ideal for deformation monitoring[7]. The system ingests each new Sentinel-1 image and processes it via ENVI SARscape or SARScape analytics toolbox. We also acquire daily rainfall data as well as forecasts and alerts from Nepal's Department of Hydrology and Meteorology (DHM). DHM already issues landslide warnings whenever upstream rainfall exceeds threshold values[8]; our system will use these forecasts (e.g. 24–48h accumulations) as one trigger input. GIS base layers will include a detailed road-network shapefile (the Narayanghat–Kathmandu highway corridor), a displacement data obtained from InSAR analysis, a digital elevation model, and any local landslide inventory. These static layers allow mapping displacements onto highway segments and identifying known hotspots.







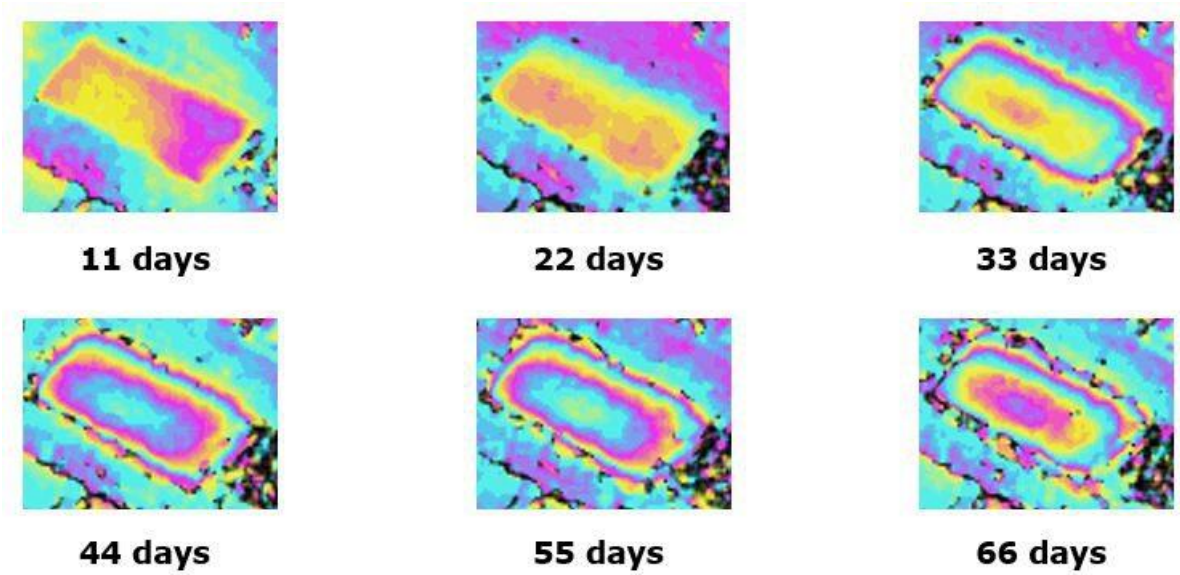


Fig2: Ground Displacement using InSAR analysis

InSAR Processing (PS/SBAS):

Sentinel-1 time series will be analyzed using two complementary InSAR approaches. Persistent Scatterer (PS-InSAR) uses a single "master" image and identifies stable radar reflectors; it excels in areas (e.g. road surfaces, rock outcrops) with little temporal decorrelation[9]. Small-Baseline Subset (SBAS) uses many short-baseline interferograms to maximize spatial coverage, especially in natural terrain[10]. Both methods will be tested in SARscape's InSAR toolbox. SARscape (ENVI SARscape) is a mature platform for interferometric stacking; for example, one national system ran fully automated PS-InSAR pipelines on cloud/GPU hardware using SARscape[11][12]. We will set up SARscape workflows (connection graph, co-registration/interferogram formation, inversion, geocoding) to generate deformation maps and time-series rasters. These workflows are activated as soon as new SAR data arrive, targeting ~24h latency[13]. All outputs (point and raster displacements) will be geocoded to the highway.

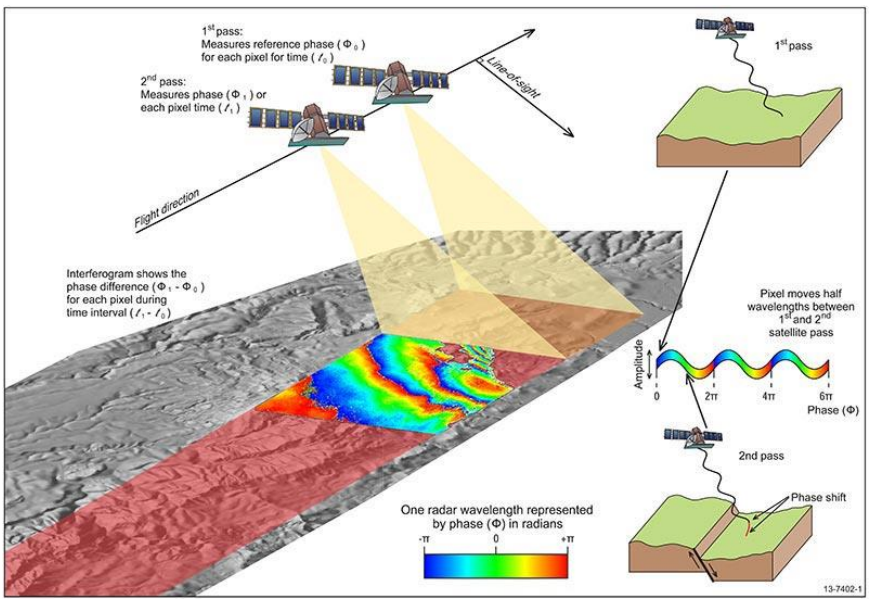


Fig3: Displacement calculation using InSAR processing

System Architecture:

The system will run on secured servers (on-premises or cloud).

Data Ingestion: Sentinel-1 scenes are downloaded daily from ESA and pre-processed (orbit correction, radiometric calibration) in SARscape. Rainfall forecasts (grid or gauge data from DHM) are retrieved via API or portal.

Processing Pipeline: SARscape executes PS and SBAS chains automatically in a sliding-window mode[12]. The pipeline exploits parallel/GPU execution for efficiency, and uses an "evolving" stack: as each new image arrives, the oldest is dropped and the stack reprocessed[14].

Analysis: A time-series analyzer tool extracts displacement versus time at key points along the route.

Risk Module: This combines SAR displacements with rainfall.

Database & UI: Results and metadata are stored in a spatial database. A Web GIS dashboard (built on open or ESRI GIS) will display current deformation maps, time plots for monitored points, rainfall warnings, and risk levels on the road. Users (NDRRMA, local DRR officers) see an easy map interface: e.g., road segments colored by InSAR velocity and rain alerts. This design follows successful precedents e.g., a Hungarian EWS automated monthly Sentinel-1 processing and delivered near-real-time warnings via a dedicated UI[13].

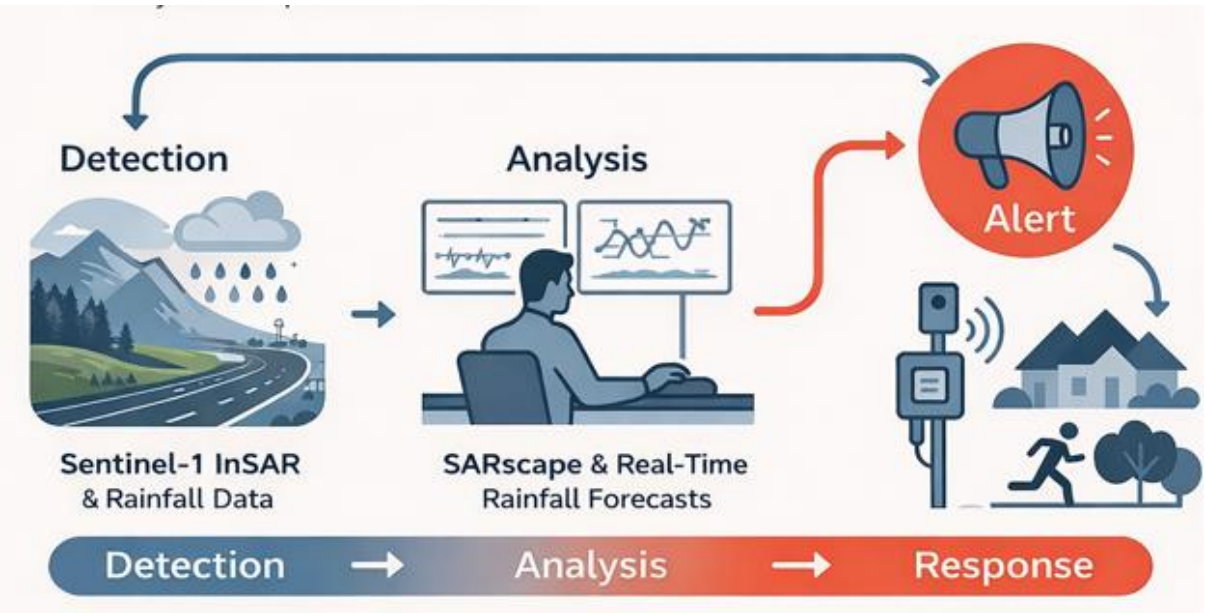


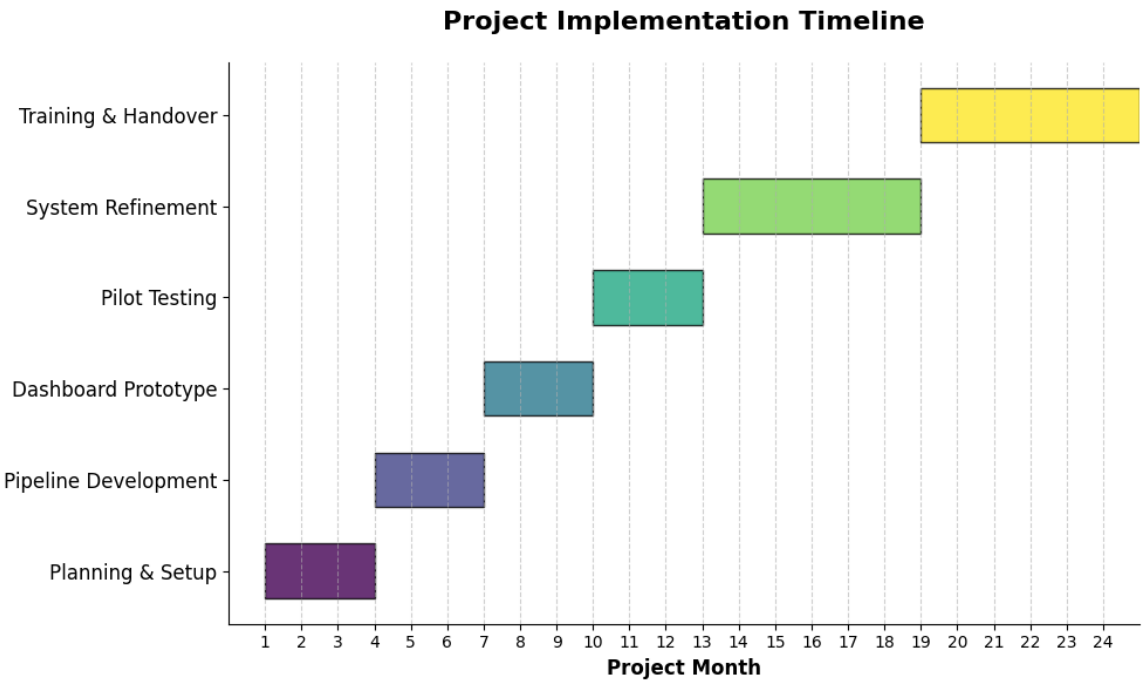
Fig4: landslide early warning system

Risk Fusion & Alert Criteria:

We will define a scoring scheme that triggers alerts when both displacement and rainfall criteria are met. In practice this might be a multi-tier warning: for example, a Level-1 (Watch) alert if either heavy rainfall or moderate creep is detected, and a Level-2 (Warning) if *both* rainfall and InSAR motion are high. Empirical thresholds will be calibrated from historical monsoon data.

Prior studies show that combining rainfall accumulation with deformation is effective: in Umbria (Italy) 48h rainfall and soil saturation thresholds closely matched InSAR-detected accelerations[15]. We will similarly compute recent (e.g. last 6–12 day) displacement rates from InSAR and compare against baseline stability (e.g. high-velocity anomalies). If the recent InSAR velocity exceeds a set threshold *and* forecasted rainfall (e.g. 48h total) exceeds the DHM critical value, the system issues a warning. These thresholds can be refined iteratively by back-testing: for example, analyzing past landslides along the highway to see what rainfall/InSAR patterns preceded failures. Since absolute LOS motion depends on viewing geometry, the system will convert LOS data to vertical/slope-normal components using the local DEM. In summary, alerts combine geophysical metrics: "Displacement Alert" when the absolute motion rate is anomalous, and "Rainfall Alert" per DHM's criteria[8] – a combined alert issues only when both are high. This fusion approach aligns with best practice in EWS design[15][5].

Implementation Plan and Timeline: We propose a two-year phased plan:



Throughout the project, progress reviews will ensure goals are met on schedule. We expect the system to ingest new Sentinel-1 data within 24 h of acquisition[13], enabling near-real-time ground motion updates.

Budget Summary (2-year estimate, USD):

Subject	**Tasks**	**Cost**
Personnel	Two full-time research staff (e.g. PI+postdoc or two specialists) for SAR/InSAR development and coordination	
Equipment	Dedicated processing hardware (workstation/servers with GPUs for SARscape), and storage	
Software	SARscape licenses (if not already provided by partner) and GIS tools.	
Travel/Workshops	Field visits in Nepal for ground validation and stakeholder training workshops.	
Overhead & Misc	Institutional overhead, data acquisition (if any), and contingency.	
Total		

This comprehensive proposal leverages the University's geospatial expertise and international partnerships to build a prototype landslide EWS tailored for Nepal. By combining *real-time* InSAR deformation data with meteorological forecasts and highway maps, the system will fill a critical gap: predicting **where and when** monsoon rains are most likely to trigger slopes to move. Early evacuation warnings based on measured deformation have proven highly effective in saving lives[5]. Our recommended approach citing precedent systems[13] ensures an operational, user-friendly EWS that empowers Nepali authorities to respond to landslide threats proactively.

Sources:

A range of recent studies and reports underpin this design. For example, NASA/ESA Sentinel-1 data provide free, six-day-revisit C-band SAR[7]; Persistent Scatterer and SBAS InSAR methods are standard for detecting slow ground motion[9][10]; and national policies emphasize multi-stakeholder EWS coordination under NDRRMA[3][4]. Landslide research confirms that short-term rainfall plus deformation thresholds reliably signal failure risks[15][5]. These references (cited above) document the technical basis and justify each element of our plan.

Work Cited

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